the most common response was 2, which is the mode. A frequency distribution that has two distinct peaks is called *bimodal*.

1.4.4 Describing Data Variability: Standard Deviation and Variance

To describe quantitative data more fully, we describe not only the center but also the *variability* about that center. The difference between the largest and smallest observations, called the ***range***, is a simple way to do this. The range incorporates only the two most extreme observations, however, so it is sensitive to outliers and does not reflect how far observations other than those two fall from the center.

More useful measures of variability use the distances of *all* the observations from the center. For observation i, the ***deviation*** of y_i from the mean $\bar{y}$ is $(y_i - \bar{y})$. The deviation is *positive* when y_i falls *above* the mean and *negative* when y_i falls *below* the mean. Calling the mean the *center of gravity* of the data reflects that the sum of the positive deviations equals the negative of the sum of negative deviations. That is, the sum of all the deviations,

$$\sum_{i=1}^{n}(y_i - \bar{y}) = \sum_{i=1}^{n} y_i - n\bar{y} = n\bar{y} - n\bar{y} = 0.$$

Because of this, measures of variability use the squares or the absolute values of the deviations.

Standard deviation and variance

For a variable y with n observations $y_1, y_2, \ldots, y_n$ in a sample from some population, the ***standard deviation*** s is

$$s = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{(y_1 - \bar{y})^2 + (y_2 - \bar{y})^2 + \cdots + (y_n - \bar{y})^2}{n-1}}.$$

The standard deviation is the positive square root of the ***variance*** s^2,

$$s^2 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2}{n-1}.$$

The variance is approximately an average of the squared deviations. The units of measurement are the squares of those for the original data, so the standard deviation is simpler to interpret: s is a sort of *typical distance* of an observation from the mean. Also, *the larger the standard deviation, the greater the spread of the data.* The value of s is nonnegative, with $s = 0$ only when all observations have the same value. The denominator of s^2 uses $(n-1)$, rather than n, because this version of the measure naturally arises in inferential statistical methods.[11]

The magnitude of s partly reflects the shape of the frequency distribution. If the distribution is approximately bell-shaped, then:[12]

1. About 68% of the observations fall between $\bar{y} - s$ and $\bar{y} + s$.
2. About 95% of the observations fall between $\bar{y} - 2s$ and $\bar{y} + 2s$.

[11] With data for an entire population, we replace $(n-1)$ by n; the variance is then precisely the mean squared deviation. In using a sample to *estimate* variability around the population mean, whose value is unknown, Section 4.4.6 shows that the bias due to the numerator of s^2 having the *sample* mean instead of the *population* mean is eliminated when we use $(n-1)$ rather than n in the denominator.

[12] Section 2.5.1 will show where these percentages come from.